# Cisco Catalyst C9200L-24P-4G Switch Manual

## 1. Official Cisco Sources

This manual is based on official Cisco documentation for the Cisco Catalyst 9200 Series Switches, especially the Cisco Catalyst 9200 Series Switches Data Sheet, the Catalyst 9200 Series overview page, and the Catalyst 9200 Series support and documentation portal.

The uploaded source manual identifies the Cisco Catalyst C9200L-24P-4G as a Cisco Catalyst 9200L fixed-uplink access switch running Cisco IOS XE. It also documents its main hardware characteristics, including twenty-four 1G RJ-45 PoE+ access ports, four fixed 1G SFP uplinks, StackWise-80 stacking support, console access, PoE budget information, software tiers, and normal troubleshooting areas.

Practical Cisco IOS XE-style configuration examples were added to make this manual more useful for real deployment, maintenance, and troubleshooting. Interface names must always be verified on the actual switch using:

```
show ip interface brief
show interfaces status
```

Because port numbering can change depending on stack member number, hardware model, software version, and actual deployment.

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## 2. Device Description

The Cisco Catalyst C9200L-24P-4G is a fixed-configuration access-layer switch from the Cisco Catalyst 9200L family. It runs Cisco IOS XE and is designed for enterprise access networks, branch offices, wiring closets, and smart office environments.

The switch provides wired network access for endpoint devices such as:

- Desktop PCs
- IP phones
- Wireless access points
- CCTV cameras
- Printers
- Smart displays
- AV control devices
- Building automation devices
- IoT or smart office devices

The model name can be understood as follows:

- **C9200L**: Cisco Catalyst 9200L fixed-uplink switch family
- **24P**: 24 Ethernet access ports with PoE+

- **4G**: 4 fixed 1G SFP uplink ports

In a smart office topology, this switch normally sits between the Cisco ISR router and the internal endpoint devices. It provides the access layer of the network, while the router provides WAN routing and internet connectivity.

The switch is suitable for:

- Access-layer switching
- VLAN segmentation
- PoE+ endpoint power
- Uplink connectivity to router, firewall, distribution switch, or core switch
- Secure wired access
- Branch network connectivity
- Centralized management through Cisco tools

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# 3. Technical Specifications

The Cisco Catalyst C9200L-24P-4G is manufactured by Cisco and belongs to the Catalyst 9200L Series.

It is a fixed access-layer Ethernet switch with Cisco IOS XE software. The switch is available with Cisco software licensing tiers such as Network Essentials and Network Advantage, depending on the ordered model and license.

The switch provides:

- 24 x 1G RJ-45 Ethernet access ports
- Full PoE+ capability on access ports
- 4 x fixed 1G SFP uplink ports
- RJ-45 console port
- Mini-USB console port
- AC power supply support
- Primary PSU and optional secondary PSU support
- StackWise-80 stacking support with the required stack kit
- 1U rack-mountable chassis
- Fixed redundant fans
- Cisco IOS XE management and operation

The default AC power supply identified for this model is commonly the Cisco 600 W AC power supply. A secondary power supply can be added for redundancy and expanded PoE power budget.

The switch supports PoE+ based on IEEE 802.3at, which can provide up to 30 W per port depending on available power budget and endpoint negotiation. The uploaded source manual states a default PoE budget of about 370 W and an expanded budget of about 740 W when a secondary power supply is installed.

The switch has:

- 2 GB DRAM
- 4 GB flash
- 56 Gbps switching capacity

- 41.67 Mpps forwarding rate
- 16,000 MAC address table entries
- StackWise-80 stacking capability

The chassis dimensions are approximately:

```
1.73 x 17.5 x 12.9 inches
```

The device is designed to be installed in a rack, wiring closet, network cabinet, or branch office communications room.

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# 4. Ports, Buttons, and Indicators

The Cisco Catalyst C9200L-24P-4G includes ports and indicators for endpoint connectivity, uplink connectivity, management, power, stacking, and physical identification.

## 4.1 Access Ports

The switch includes twenty-four RJ-45 Gigabit Ethernet access ports.

These ports are used to connect:

- PCs
- IP phones
- Access points
- CCTV cameras
- Printers
- Smart TVs or commercial displays
- Meeting room devices
- AV controllers
- Other wired endpoint devices

Each access port can provide PoE+ power if the connected device supports it and if enough PoE budget is available.

Typical access port use:

```
Endpoint device → RJ-45 access port → Catalyst switch → uplink → router/core
```

## 4.2 SFP Uplink Ports

The switch includes four fixed 1G SFP uplink ports.

These uplinks are normally used to connect the switch to:

- Router

- Firewall
- Distribution switch
- Core switch
- Another access switch
- Fiber or copper SFP connection

The "4G" in the model name refers to the four fixed 1G uplink ports.

Only Cisco-supported or compatible SFP modules should be used, and SFP support should be verified from Cisco documentation before deployment.

### 4.3 Console Ports

The switch provides local management access through:

- RJ-45 console port
- Mini-USB console port

Console access is used for:

- First-time configuration
- Password recovery
- Troubleshooting management access
- Recovery when SSH or remote access fails
- Viewing boot messages

### 4.4 Power Supply Bays

The switch supports a primary AC power supply and can support a secondary power supply depending on configuration.

The secondary power supply can provide:

- Power redundancy
- Higher PoE budget
- Better availability

If one PSU fails and a secondary PSU is installed, the switch may continue operating depending on power load and configuration.

### 4.5 StackWise Ports

The switch supports StackWise-80 stacking, but stacking requires the proper C9200L stack kit and stack cables.

Stacking allows multiple compatible switches to operate as one logical switch.

Stacking is useful for:

- Expanding port count
- Centralized management
- Simplified access-layer design
- Redundancy between access switches

### 4.6 LEDs

The switch includes LEDs for:

- System status
- Port link/activity
- PoE status
- Uplink activity
- Power supply status
- Beacon identification

The blue beacon LED helps physically identify the switch in a rack or network cabinet.

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## 5. Installation and Setup Notes

Before installing the switch, confirm the exact model, power supply, rack accessories, stack kit requirements, uplink SFPs, and endpoint cabling.

Basic installation steps:

1. Install the switch in a 1U rack position or secure network cabinet.
2. Connect the Cisco AC power supply.
3. Add a secondary PSU if redundancy or extra PoE budget is needed.
4. Connect access ports to endpoint devices.
5. Connect SFP uplinks to the router, firewall, distribution switch, or core switch.
6. Connect the console cable for first-time setup.
7. Verify boot status through LEDs and console output.
8. Configure hostname, management IP, VLANs, uplinks, and security.
9. Save the configuration.
10. Label ports and document the final connections.

The switch should be installed in an environment with proper airflow. Do not block the fan exhaust or ventilation areas. Keep the switch away from dust, moisture, excessive heat, and unstable power sources.

For PoE-heavy deployments, calculate the expected power draw before connecting all devices. Devices such as access points, cameras, and IP phones may consume significant PoE power.

---

## 6. Basic Initial Configuration

The following commands show a practical IOS XE-style setup for the switch.

First, access privileged mode and enter configuration mode:

```
enable
configure terminal
```

### 6.1 Set Hostname

```
hostname C9200L-SW1
```

The hostname helps identify the switch in CLI sessions, logs, monitoring tools, and topology documentation.

### 6.2 Set Enable Secret

```
enable secret StrongPasswordHere
```

The enable secret protects privileged EXEC mode.

### 6.3 Disable Unwanted DNS Lookup

```
no ip domain-lookup
```

This prevents the switch from trying to resolve mistyped CLI commands as domain names.

### 6.4 Create Local Admin User

```
username admin privilege 15 secret StrongAdminPassword
```

This creates a local administrator account for secure access.

### 6.5 Save Configuration

```
copy running-config startup-config
```

This step is critical. If changes are not saved, they may be lost after reload or power failure.

---

## 7. Management IP Configuration

A switch needs a management IP address so administrators can access it remotely through SSH, monitoring tools, or Cisco management platforms.

Management IP is usually configured on an SVI, such as VLAN 10 or VLAN 99.

Example:

```
configure terminal
vlan 99
 name MANAGEMENT
```

```
exit

interface vlan 99
 description Management SVI
 ip address 192.168.99.2 255.255.255.0
 no shutdown
exit
```

If the switch is operating as a Layer 2 access switch, configure a default gateway:

```
ip default-gateway 192.168.99.1
```

If Layer 3 routing is enabled on the switch, use a default route instead:

```
ip routing
ip route 0.0.0.0 0.0.0.0 192.168.99.1
```

Verify:

```
show ip interface brief
ping 192.168.99.1
```

Expected result:

- VLAN interface is up/up
- Management IP is reachable
- Default gateway or default route is correct

Important note:

The management SVI will only come up if the VLAN exists and at least one active port in that VLAN is up, or if the VLAN is active through a trunk.

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# 8. SSH and CLI Access Configuration

SSH should be used for secure remote access. Telnet should be avoided.

## 8.1 Configure SSH

```
configure terminal
ip domain-name smartoffice.local
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
```

```
 login local
 transport input ssh
 exec-timeout 5 30
exit
```

### 8.2 Verify SSH

```
show ip ssh
show running-config | section line vty
```

If SSH fails, check:

- Management IP reachability
- Username and password
- RSA keys
- Domain name
- VTY configuration
- ACL restrictions
- VLAN and gateway configuration

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## 9. VLAN Configuration

VLANs allow the switch to separate traffic logically.

Common smart office VLAN examples:

```
VLAN 10  - Users
VLAN 20  - Voice
VLAN 30  - Wi-Fi / Access Points
VLAN 40  - CCTV
VLAN 50  - AV / Meeting Room Devices
VLAN 99  - Management
```

Create VLANs:

```
configure terminal
vlan 10
 name USERS
vlan 20
 name VOICE
vlan 30
 name WIFI_APS
vlan 40
 name CCTV
vlan 50
```

```
 name AV_DEVICES
vlan 99
 name MANAGEMENT
exit
```

Verify:

```
show vlan brief
```

Expected result:

- VLANs appear in the VLAN table
- Ports can be assigned to the correct VLANs
- Trunks allow the required VLANs

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# 10. Access Port Configuration

Access ports connect endpoint devices to a single VLAN.

Example for a PC port:

```
configure terminal
interface GigabitEthernet1/0/1
 description User PC
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit
```

Example for an IP phone and PC sharing the same port:

```
configure terminal
interface GigabitEthernet1/0/2
 description IP Phone + User PC
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 20
 spanning-tree portfast
 no shutdown
exit
```

Example for CCTV camera:

```
configure terminal
interface GigabitEthernet1/0/3
 description CCTV Camera
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 no shutdown
exit
```

Verify:

```
show interfaces status
show vlan brief
show running-config interface GigabitEthernet1/0/1
```

Expected result:

- Port is connected
- Port is assigned to the correct VLAN
- Endpoint receives network connectivity
- PoE device receives power if required

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# 11. Trunk / Uplink Configuration

Trunk ports carry multiple VLANs between network devices.

A trunk uplink may connect the C9200L switch to:

- Router
- Firewall
- Distribution switch
- Core switch
- Another switch

Example trunk configuration:

```
configure terminal
interface GigabitEthernet1/1/1
 description Uplink to Router or Core Switch
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,99
 no shutdown
exit
```

Verify trunk status:

```
show interfaces trunk
show interfaces status
show spanning-tree interface GigabitEthernet1/1/1
```

Expected result:

- Interface is up/up
- Trunk is active
- Required VLANs are allowed
- No unexpected VLAN is missing
- Spanning Tree is not blocking the uplink unless expected

If the uplink does not work, check:

- SFP compatibility
- Fiber/copper cable
- Allowed VLAN list
- Native VLAN mismatch
- Speed/duplex negotiation
- Remote-side configuration
- Spanning Tree state

---

## 12. PoE and PoE+ Configuration

The C9200L-24P-4G supports PoE+ on its access ports. PoE is used to power devices such as:

- IP phones
- Wireless access points
- CCTV cameras
- Door controllers
- Some AV devices
- IoT devices

### 12.1 Check PoE Status

```
show power inline
```

This command shows:

- Available PoE power
- Used PoE power
- Port-by-port power status
- Device power draw
- PoE class information

### 12.2 Enable PoE on a Port

PoE is usually enabled automatically, but it can be controlled manually.

```
configure terminal
interface GigabitEthernet1/0/5
 power inline auto
exit
```

### 12.3 Disable PoE on a Port

```
configure terminal
interface GigabitEthernet1/0/5
 power inline never
exit
```

### 12.4 Troubleshooting PoE

If a device does not power on:

```
show power inline
show interfaces status
show logging
show running-config interface GigabitEthernet1/0/5
```

Possible causes:

- PoE budget exceeded
- Device requires more power than available
- Bad cable
- Port disabled
- PoE disabled on port
- Endpoint does not support standard PoE
- Faulty endpoint
- Faulty switch port
- Insufficient power supply capacity

Fix actions:

- Move the device to another port
- Replace the cable
- Enable PoE with `power inline auto`
- Check total PoE budget
- Add a secondary PSU if more PoE budget is needed
- Confirm endpoint power requirements

## 13. StackWise-80 Stacking

The C9200L supports StackWise-80 stacking when the proper stack kit and cables are installed.

Stacking allows multiple switches to operate as one logical switch.

Benefits:

- One management plane
- Higher port count
- Easier configuration
- Simplified access-layer design
- Better redundancy

Verify stack status:

```
show switch
show switch stack-ports
```

Expected result:

- All members appear in the stack
- One switch is active
- Other members appear as standby/member
- Stack ports are up
- Stack cables are detected

If a stack member does not join:

- Check stack kit installation
- Check stack cables
- Check switch compatibility
- Check IOS XE version compatibility
- Check stack member priority
- Reload if required during maintenance window

Stacking should be planned before production deployment because adding or removing stack members may require downtime.

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## 14. Normal Operation

During normal operation:

- System LED indicates healthy switch operation
- Power supply LEDs show proper power state
- Access port LEDs show link and activity
- PoE LEDs indicate power delivery status

- Uplink ports are up and passing traffic
- VLANs are correctly assigned
- Trunks carry required VLANs
- Management IP is reachable
- SSH access works
- PoE devices receive power
- Logs do not show repeated hardware, power, or link errors

Useful verification commands:

```
show version
show inventory
show ip interface brief
show interfaces status
show vlan brief
show interfaces trunk
show power inline
show logging
show running-config
show startup-config
show mac address-table
show spanning-tree
```

---

# 15. Troubleshooting

Troubleshooting should start from the physical layer and move upward:

1. Power
2. Cabling
3. Port state
4. VLAN assignment
5. Trunk configuration
6. PoE budget
7. Spanning Tree
8. Management IP
9. Software image
10. Hardware fault

---

## 15.1 Switch Does Not Power On

Symptom:

- No LEDs
- No fan activity
- No console output
- Switch does not boot

Possible causes:

- Power cable disconnected
- Failed power outlet
- Faulty PSU
- PSU not seated correctly
- Wrong power source
- Internal hardware failure

Checks:

```
1. Confirm the power cable is connected.
2. Test the outlet with another device.
3. Reseat the power supply.
4. Check PSU LED.
5. Try another supported PSU if available.
6. Connect through console and check boot output.
```

Resolution:

If the switch remains completely off after testing with a known-good power source and supported PSU, escalate for hardware replacement or Cisco support.

---

### 15.2 Access Port Is Down

Symptom:

- Endpoint has no network connection
- Port LED is off
- `show interfaces status` shows notconnect or disabled

Useful commands:

```
show interfaces status
show ip interface brief
show running-config interface GigabitEthernet1/0/1
show interfaces GigabitEthernet1/0/1
```

Possible causes:

- Bad cable
- Endpoint powered off
- Port administratively shut down
- Wrong port used
- Port security violation
- VLAN issue
- Hardware port failure

Fix:

```
configure terminal
interface GigabitEthernet1/0/1
 no shutdown
exit
```

Then replace the cable and test with another endpoint.

---

### 15.3 Endpoint Connected but No Network Access

Symptom:

- Port is physically up
- Endpoint cannot reach gateway
- User cannot access network resources

Possible causes:

- Wrong access VLAN
- Missing VLAN
- Trunk not carrying VLAN
- DHCP issue
- Gateway issue
- ACL/security policy issue
- STP blocking path

Useful commands:

```
show vlan brief
show interfaces trunk
show mac address-table interface GigabitEthernet1/0/1
show spanning-tree vlan 10
show running-config interface GigabitEthernet1/0/1
```

Fix actions:

- Assign correct VLAN
- Create missing VLAN
- Allow VLAN on trunk
- Verify gateway availability
- Check DHCP server or relay
- Check security policy

Example:

```
configure terminal
interface GigabitEthernet1/0/1
 switchport mode access
 switchport access vlan 10
 no shutdown
exit
```

---

### 15.4 PoE Device Does Not Power On

Symptom:

- IP phone, access point, or camera does not power up
- Port may show link down
- `show power inline` shows denied, faulty, or no power

Useful commands:

```
show power inline
show power inline GigabitEthernet1/0/5
show interfaces status
show logging
```

Possible causes:

- PoE budget exceeded
- Port PoE disabled
- Bad cable
- Endpoint requires more power
- Faulty endpoint
- Faulty switch port
- Insufficient PSU capacity

Fix:

```
configure terminal
interface GigabitEthernet1/0/5
 power inline auto
 no shutdown
exit
```

If the power budget is exceeded, remove unnecessary PoE load or add a secondary PSU if supported and required.

---

### 15.5 Uplink Connectivity Problem

Symptom:

- Switch access ports are up
- Users cannot reach router, core, or internet
- SFP uplink is down or unstable

Useful commands:

```
show interfaces status
show interfaces trunk
show interfaces GigabitEthernet1/1/1
show logging
show inventory
```

Possible causes:

- Unsupported SFP
- Bad fiber or copper cable
- Remote-side interface down
- Trunk mismatch
- Native VLAN mismatch
- Allowed VLAN list missing required VLANs
- Speed/duplex issue
- Spanning Tree blocking uplink

Fix actions:

- Reseat SFP
- Replace fiber/copper cable
- Verify supported SFP
- Match trunk configuration on both sides
- Allow required VLANs
- Check remote device port
- Review Spanning Tree status

Example trunk fix:

```
configure terminal
interface GigabitEthernet1/1/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,99
 no shutdown
exit
```

---

## 15.6 VLAN or Trunk Problem

Symptom:

- Some VLANs work while others fail
- Devices in one VLAN cannot reach gateway
- Management VLAN unreachable
- Phones or cameras fail while PCs work

Useful commands:

```
show vlan brief
show interfaces trunk
show spanning-tree
show running-config interface GigabitEthernet1/1/1
```

Possible causes:

- VLAN not created
- VLAN not allowed on trunk
- Wrong access VLAN
- Native VLAN mismatch
- Trunk mode not enabled
- STP blocking the VLAN

Fix:

```
configure terminal
vlan 10
 name USERS
exit

interface GigabitEthernet1/1/1
 switchport mode trunk
 switchport trunk allowed vlan add 10
exit
```

---

## 15.7 DHCP Problem for Connected Devices

Symptom:

- Devices receive 169.254.x.x addresses
- Devices have no default gateway
- Devices cannot access network

Possible causes:

- DHCP server unreachable
- VLAN not allowed on trunk

- Wrong access VLAN
- Gateway/SVI down
- DHCP relay missing on router or Layer 3 device
- DHCP scope exhausted

Useful switch commands:

```
show vlan brief
show interfaces trunk
show mac address-table
show spanning-tree vlan 10
```

Switch-side fix actions:

- Confirm endpoint port VLAN
- Confirm VLAN exists
- Confirm uplink trunk carries VLAN
- Confirm uplink is not blocked
- Confirm gateway/router is reachable

---

### 15.8 Management Access Failure

Symptom:

- Switch is forwarding traffic
- Admin cannot SSH into switch
- Management IP does not respond

Useful commands from console:

```
show ip interface brief
show running-config | section line vty
show ip ssh
show users
show access-lists
show vlan brief
show interfaces trunk
```

Possible causes:

- Management SVI down
- Wrong management IP
- Wrong default gateway
- SSH not configured
- RSA keys missing
- VTY line misconfigured

- ACL blocks admin subnet
- Management VLAN not carried on trunk

Fix example:

```
configure terminal
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
exit
ip default-gateway 192.168.99.1
```

SSH fix:

```
configure terminal
ip domain-name smartoffice.local
username admin privilege 15 secret StrongAdminPassword
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
exit
```

---

### 15.9 Spanning Tree Blocking or Loop Issue

Symptom:

- Network instability
- Some ports stop forwarding
- Intermittent connectivity
- MAC address flapping
- High broadcast traffic

Useful commands:

```
show spanning-tree
show spanning-tree vlan 10
show logging
show mac address-table dynamic
```

Possible causes:

- Network loop
- Wrong cabling between switches

- Missing STP planning
- Incorrect trunk design
- Endpoint connected through unmanaged switch
- Portfast used incorrectly on switch-to-switch link

Fix actions:

- Remove looped cable
- Check uplink design
- Do not use PortFast on switch-to-switch trunks
- Use PortFast only on endpoint access ports
- Review STP root bridge placement

---

### 15.10 High Interface Errors

Symptom:

- Slow network
- Packet loss
- CRC errors
- Input errors
- Drops on interface

Useful commands:

```
show interfaces GigabitEthernet1/0/1
show interfaces counters errors
show logging
```

Possible causes:

- Bad cable
- Bad patch panel
- Duplex/speed mismatch
- Faulty endpoint NIC
- Faulty SFP
- Dirty fiber connector
- Physical layer issue

Fix actions:

- Replace patch cable
- Test another switch port
- Test another endpoint
- Clean fiber connectors
- Replace SFP if needed
- Verify speed and duplex negotiation

---

### 15.11 Stack Member Does Not Join

Symptom:

- Stack member missing
- Switch operates standalone
- Stack ports down
- Stack LED/status abnormal

Useful commands:

```
show switch
show switch stack-ports
show version
show logging
```

Possible causes:

- Stack kit missing
- Stack cable disconnected
- Stack cable faulty
- IOS XE mismatch
- Stack member priority issue
- Unsupported stack combination
- Switch not reloaded after stacking changes

Fix actions:

- Install proper C9200L stack kit
- Reseat stack cables
- Verify stack port status
- Match IOS XE versions
- Reload during maintenance window if needed
- Confirm hardware compatibility

---

### 15.12 Software or Boot Issue

Symptom:

- Switch fails to boot
- Switch reloads repeatedly
- IOS XE image appears corrupted
- Console shows boot errors

Useful commands if accessible:

```
show version
show flash:
show boot
dir flash:
```

Possible causes:

- Corrupt IOS XE image
- Unsupported software image
- Interrupted upgrade
- Flash storage issue
- Wrong boot variable

Fix actions:

- Use console access
- Verify boot variable
- Load supported IOS XE image
- Check flash storage
- Follow Cisco recovery procedure if normal boot fails

---

### 15.13 Configuration Lost After Reload

Symptom:

- Switch comes back with old or blank configuration
- Recent changes disappear
- VLANs or interfaces return to previous state

Possible causes:

- Running configuration was not saved
- Startup configuration was erased
- Factory reset was performed
- Wrong boot or recovery action

Fix:

```
copy running-config startup-config
```

Verify:

```
show startup-config
show running-config
```

Best practice:

Always save the configuration after changes and keep an external backup.

---

## 16. Factory Reset and Recovery

Factory reset should be used carefully because it removes the saved configuration.

A common reset method is:

```
write erase
reload
```

Alternative command depending on software version:

```
erase startup-config
reload
```

Before reset, back up the configuration if possible:

```
copy running-config startup-config
copy running-config flash:backup-config.cfg
```

You can also copy the configuration to an external server if available:

```
copy running-config tftp:
```

Factory reset is used when:

- The switch is being repurposed
- Configuration is corrupted
- Admin access is lost and recovery is required
- A clean setup is needed

Important warning:

After reset, VLANs, management IP, SSH, port settings, trunks, PoE settings, and all custom configuration may be removed. The switch must be reconfigured before returning to production.

---

## 17. Maintenance and Best Practices

Recommended practices:

- Keep Cisco IOS XE on a supported release.
- Back up the running configuration after changes.
- Document port assignments.
- Label access and uplink cables.
- Monitor PoE budget.
- Use supported SFP modules.
- Keep spare patch cables and SFPs.
- Use SSH instead of Telnet.
- Restrict management access.
- Keep the switch in a ventilated rack.
- Keep vents and fans clear.
- Monitor logs regularly.
- Check interface errors periodically.
- Avoid unnecessary reloads.
- Use graceful reload when possible.
- Plan stacking changes during maintenance windows.
- Keep a copy of the switch configuration outside the device.

Useful maintenance commands:

```
show version
show inventory
show environment
show logging
show interfaces status
show interfaces counters errors
show power inline
show vlan brief
show interfaces trunk
show running-config
show startup-config
copy running-config startup-config
```

---

# 18. Quick Reference

## Power Check

```
Check power cable
Check power outlet
Check PSU LED
Check system LED
Check console boot output
```

## Interface Check

```
show interfaces status
show ip interface brief
show interfaces
```

### VLAN Check

```
show vlan brief
show running-config | section vlan
```

### Trunk Check

```
show interfaces trunk
show running-config interface GigabitEthernet1/1/1
```

### PoE Check

```
show power inline
show power inline GigabitEthernet1/0/5
```

### Management Check

```
show ip interface brief
show ip ssh
show running-config | section line vty
ping <default-gateway>
```

### Spanning Tree Check

```
show spanning-tree
show spanning-tree vlan <vlan-id>
```

### Stack Check

```
show switch
show switch stack-ports
```

### Save Configuration

```
copy running-config startup-config
```

### Access Port Setup

```
interface GigabitEthernet1/0/1
 description User Device
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
```

### Voice Port Setup

```
interface GigabitEthernet1/0/2
 description IP Phone and PC
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 20
 spanning-tree portfast
 no shutdown
```

### Trunk Uplink Setup

```
interface GigabitEthernet1/1/1
 description Uplink
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,99
 no shutdown
```

### Management SVI Setup

```
vlan 99
 name MANAGEMENT
exit

interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
exit

ip default-gateway 192.168.99.1
```

## SSH Setup

```
ip domain-name smartoffice.local
username admin privilege 15 secret <PASSWORD>
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
```

## Factory Reset

```
write erase
reload
```